Yujie Zang

Email: zangyujie@u.nus.sg | Homepage: https://zangyujie2004.github.io/ | Scholar

EDUCATION

National University of Singapore *Singapore*

M.Sc. in Robotics *Jul. 2026 – Present*

Selected Coursework: Robot Kinematics, Robot Dynamics and Control, AI Planning and Decision Making, Machine Learning in Robotics

Shanghai Jiao Tong University *Shanghai, China*

B.Eng. in Mechanical Industrial Design (Outstanding Graduate) *Aug. 2022 – Jun. 2026*

Grades: 90.10 / 100 | GPA: 3.91 / 4.30 | Ranking: 4 / 36

Selected Coursework: Product Design, Materials and Structures, Advanced Mathematics, Computer Programming, Data Science, Machine Learning

RESEARCH INTEREST

My long-term research goal is to build robotic systems that leverage **human priors** to advance toward **superhuman physical intelligence**. In particular, I focus on **real-world robotic manipulation**, using **force and tactile sensing** to understand physical interactions, **predictive model** to predict their evolution, and **real-robot post-training** to acquire precise and fast skills.

PUBLICATIONS

(† equal contribution; * corresponding author)

[1] TacForeSight: Force-Guided Tactile World Model for Contact-Rich Manipulation. ***[Project Page]***

Yujie Zang†, Yuhang Zheng†, Xian Nie†, Yupeng Zheng, Shuai Tian, Songen Gu, Chen Gao, Zining Wang, Shuicheng Yan*, Wenchao Ding*.

IEEE Robotics and Automation Letters, 2026. (Under Review)

[2] Interaction Latents As Actions for Contact-Rich Manipulation. ***[Project Page]***

Yuhang Zheng†, **Yujie Zang**†, Yuchen Liu†, Yupeng Zheng, Zhennan Jiang, Zebin Xing, Shuai Tian, Songen Gu, Jingxiang Guo, Chen Gao, Zhongpu Xia*, Shuicheng Yan*.

International Conference on Learning Representations, 2027. (Under Review)

[3] Contact Trajectory Prompting: In-Context Transfer of Contact-Rich Behaviors from a Single Demonstration. ***[Project Page]***

Xian Nie†, **Yujie Zang**†, Yuhang Zheng†, Yupeng Zheng, Wendi Chen, Chuan Wen, Cewu Lu, Wenhao Ding, Junchi Yan*, Shuicheng Yan*.

IEEE International Conference on Robotics and Automation, 2027. (Under Review)

[4] OmniVTA: Visuo-Tactile World Modeling for Contact-Rich Robot Manipulation. ***[Project Page]***
(EAI-100 Top 10 Open-Source Embodied AI Dataset, 2025)

Yuhang Zheng, Songen Gu, Weize Li, Yupeng Zheng, **Yujie Zang**, Shuai Tian, Xiang Li, Ruihai Wu, Ce Hao, Chen Gao, Si Liu, Haoran Li, Yilun Chen, Shuicheng Yan*, Wenchao Ding*.

IEEE Transactions on Robotics, 2026. (Under Review)

[5] VT-WAM: Visual-Tactile World Action Model for Contact-Rich Manipulation. ***[Project Page]***

Shuai Tian, Yupeng Zheng, Yuhang Zheng, Songen Gu, **Yujie Zang**, Yuxing Qin, Weize Li, Haoran Li, Wenchao Ding*, Dongbin Zhao*.

IEEE Robotics and Automation Letters, 2026. (Under Review)

[6] Multimodal Palpation Sensing for Precise Prediction of Breast Lump Hardness and Size. ***[Paper Link]***

Yujie Zang, Shuhong Xiao, Peng Yang, Ying Yang, Zixuan Lu, Ruyue Sun, Chongpeng Wei, Shuyue Zheng, Chen Wang, Xinyang Tan*, Hongwei Zhang*.

IEEE International Conference on Acoustics, Speech and Signal Processing, 2026. (Accepted)

[7] Re-DirectExo: A Preliminary Study of Illusion-Driven Hand Rehabilitation Exoskeleton. ***[Paper Link]***

Yujie Zang, Guiqin Lu, Zhaoyi Chen, Zhen Han, Zihang Ma, Xinyang Tan*, Shi Qiu*.

International Conference on Human-Engaged Computing, 2025. (Accepted)

[8] GAEA-Net: Generating Activity-Enriched Abnormal ECGs via Adversarial Network. ***[Paper Link]***

Liuqing Chen, Shuhong Xiao, **Yujie Zang**, Jiner Wang, Shanhai Hu, Lin Zhang, Qing Li, Danyang Zhang.

IEEE Journal of Biomedical and Health Informatics, 2026. (Accepted)

RESEARCH EXPERIENCE

LV-Robotics Lab, National University of Singapore *Singapore*

Research Intern | Advisor: **Prof. Shuicheng Yan** *Aug. 2026 – Present*

Topics: Contact-rich manipulation; tactile sensing and robot learning

Intelligent Medical Design Lab, Shanghai Jiao Tong University *Shanghai, China*

Research Intern | Advisors: **Prof. Guangye Li, Prof. Xinyang Tan** *Apr. 2023 – Jun. 2026*

Topics: EEG decoding and wearable sleep monitoring product development; medical hardware devices (hand exoskeleton, tumour detection product, scoliosis orthoses design)

Institute for AI Industry Research (AIR), Tsinghua University *Beijing, China*

Research Intern | Advisors: **Prof. Wenchao Ding** *Jan. 2025 – Mar. 2025*

Topics: Precious Manipulation and Robot Understanding. Evaluated whether robotic agents understand complex 3D environments.

D3 Lab, College of Computer Science and Technology, Zhejiang University *Hangzhou, China*

Research Intern | Advisor: **Prof. Liuqing Chen** *Nov. 2024 – Apr. 2025*

Topics: Activity-enriched ECG synthesis; GAN modeling for ECG signals

INDUSTRY EXPERIENCE

Romoya Robotics *Singapore*

Research Assistant | Advisor: **Prof. Shuicheng Yan** *Aug. 2026 – Present*

Topics: Robot System Development, Robot Research Assistant

Anyverse Dynamics (无界动力) *Beijing, China*

Robotics Algorithm Research Intern | Advisor: **Prof. Zhongpu Xia** *Jun. 2026 – Aug. 2026*

Topics: Dual Arm Manipulation Pipeline, Human-in-the-Loop Robot Learning, Real Robot Reinforcement Learning

TARS Robotics (它石智航) *Shanghai, China*

Robotics Algorithm Research Intern | Advisor: **Prof. Wenchao Ding** *Jan. 2026 – Jun. 2026*

Topics: Tactile Manipulation, Robot Learning, Tactile UMI Hardware Design

AWARDS

Champion, RoboMaster University Championship *Sep. 2024*

Third Prize, China–U.S. Young Maker Competition (Shanghai Division) *Jul. 2024*

Honorable Mention, Mathematical Contest in Modeling (MCM) *May 2024*

SKILLS

Robotics & Hardware: Mechanical design (SolidWorks, CAD, machining and assembly), embedded electronics, PCB and sensor integration, ROS/ROS2, robot kinematics, real-robot deployment, and ergonomic prototyping.

Robot Learning & Programming: Reinforcement learning, imitation learning, diffusion policy, tactile manipulation, and human-in-the-loop learning. Python, MATLAB, C/C++, Git; Unity, Blender, LaTeX, and Figma.

Data Analysis & Languages: Signal processing, machine learning, deep learning, time-series analysis, and data visualization. Mandarin (Native), English (Fluent).